

Solid Mechanics (EMCH260 Section 002, Fall 2021)

Mid-Term Exam 1, 8:30 AM, Wednesday, Sept. 15th 2021

Name (First/Last): _____ Score: _____/100'

1. Single choice questions. There is one and only one correct answer for each question. (10 × 3' = 30' in total)

Q1: Which one is not the unit of stress?

- A. N/m²
- B. Pa
- C. MPa
- D. kN

Correct Answer: D. kN

Explanation: The stress is defined as force divided by area, so the unit of stress is [N/m²] or [Pa] or [MPa].

Q2: Under external loadings, the internal resultant shear force happens inside a section plane and the internal resultant normal force happens in the direction that is perpendicular to that section plane.

Correct Answer: Right

Q3: The Poisson's ratio of any materials is always positive.

Correct Answer: Wrong

Explanation: The Poisson's ratio can be negative and such materials are called auxetic materials.

Q4: The Poisson's ratio connects the lateral strain and the longitudinal strain of a material under external axial or longitudinal force.

Correct Answer: Right

Q5: A material that can be subjected to very large strains before it fractures is called a brittle material.

Correct Answer: Wrong

Explanation: The *ductile* material usually undergoes a very large strain before it fractures. In contrast, the *brittle* material usually does not have large plastic deformation before it fractures.

Q6: Significant necking could happen in the plastic deformation range of ductile materials.

Correct Answer: Right

Q7: Which mechanical property is defined by the slope of the stress-strain diagram in the elastic range?

- A. Internal Resultant Loading
- B. Yield Strength
- C. Poisson's Ratio
- D. Modulus of Elasticity

Correct Answer: D. Modulus of Elasticity

Explanation: Modulus of Elasticity or Young's Modulus is defined as the slope of the stress-strain diagram in the elastic range, i.e. $E = \frac{\sigma}{\epsilon}$.

Q8: During the loading, when the stress in the material exceeds Yield Strength, the material goes into plastic deformation range.

Correct Answer: Right

Q9: What is called by an irreversible deformation that remains in a structure or material after the external loading has been fully removed?

- A. Internal Resultant Loading
- B. Permanent Set
- C. Modulus of Elasticity
- D. Yield Strength

Correct Answer: B. Permanent Set

Explanation: Permanent Set also means the plastic deformation which is the irreversible deformation in a material even after the external loading has been completely removed. Yield Strength is the point on the stress-strain diagram after which the material starts to undergo the plastic deformation range.

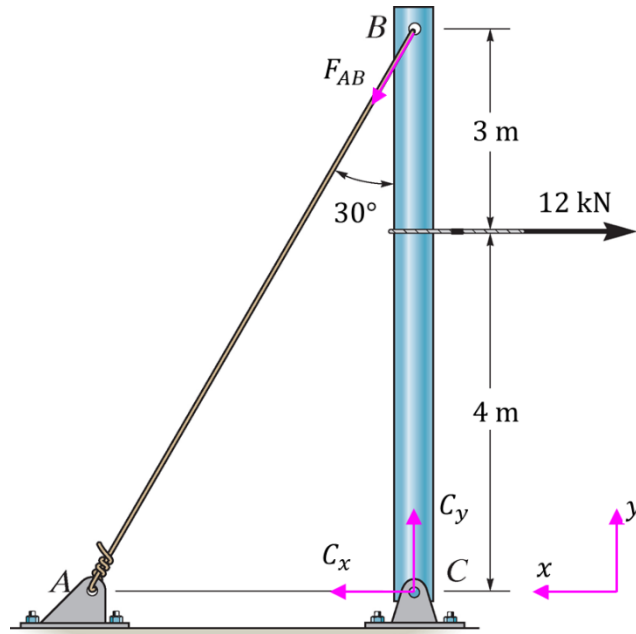
Q10: Which one is not the unit of Yield Strength?

- A. Pa
- B. MPa
- C. kN
- D. kN/m²

Correct Answer: C. kN

Explanation: Yield Strength is the point on the stress-strain diagram after which the material starts to undergo the plastic deformation range. Thus, the unit of Yield Strength is the same as stress, i.e. [N/m²] or [Pa] or [MPa].

2. The rigid pole is supported by a pin at C and a wire AB . If the wire has a diameter of 0.5 cm and its modulus of elasticity is 200 GPa, determine how much it stretches when a horizontal force of 12 kN acts on the pole and the new diameter of the wire. Assume the Poisson's ratio of the wire is 0.32. (30' in total)



Solutions:

Apply Free-Body-Diagram to the rigid pole:

$$\sum M_C = 0 \quad \rightarrow \quad F_{AB} \times L_{BC} \sin 30^\circ = 12 \times 4 = 48 \text{ kN} \cdot \text{m} \quad \rightarrow \quad F_{AB} = 13.714286 \text{ kN}$$

The tensile stress in the wire AB can be calculated as

$$\sigma_{AB} = \frac{F_{AB}}{A_{AB}} = \frac{13.714286 \times 1000}{\frac{\pi}{4} \times 0.005^2} = 698.4628 \text{ MPa}$$

Using the Hooke's law, we get the tensile (longitudinal) strain in the wire AB :

$$\varepsilon_{AB} = \frac{\sigma_{AB}}{E} = \frac{698.4628 \times 10^6}{200 \times 10^9} = 0.003492 \text{ cm/cm}$$

Using the definition of tensile strain, we then know the stretching of the wire:

$$\delta_{AB} = \varepsilon_{AB} \times L_{AB} = 0.003492 \times \frac{7}{\cos 30^\circ} = 0.02822 \text{ m} = \boxed{2.822 \text{ cm}}$$

The lateral strain can be calculated by using the Poisson's ratio:

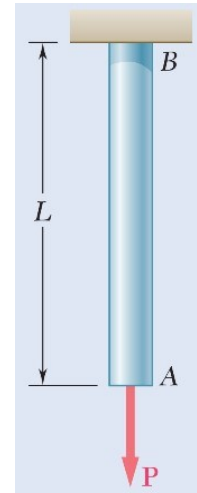
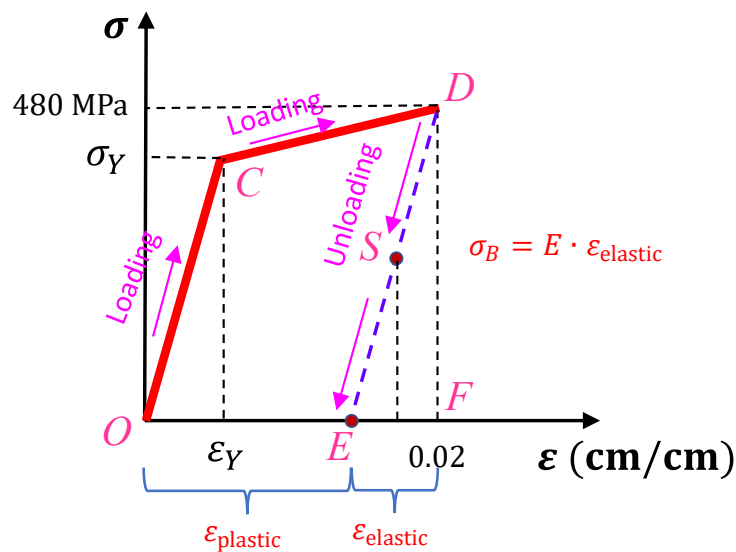
$$\varepsilon_{\text{lat}} = -\nu \varepsilon_{\text{long}} = -0.32 \times 0.003492 = -0.00111744$$

Using the definition of strain, the new diameter is

$$D = (1 + \varepsilon_{\text{lat}}) \times D_0 = (1 - 0.00111744) \times 0.5 = \boxed{0.49944 \text{ cm}}$$

3. A cylindrical rod AB has a length $L = 1$ m and a diameter of $D = 2$ cm. The Modulus of Elasticity is $E = 250$ GPa and Yield Strength is $\sigma_Y = 400$ MPa. The stress-strain diagram of the rod is shown in the left figure. The rod is subjected to an axial load P until it is stretched 2 cm. (40' in total)

- (a) Determine the external force P . (10')
 (b) After the rod is stretched, if the external force P is reduced to half of its original value, determine the total length of the rod. (20')
 (c) Continue the process in (b): if the external force P is completely removed, determine the final total length of the rod. (10')



Solutions:

- (a) According to the stretched length, the total strain induced by the external load P is:

$$\epsilon_{\text{total}} = \frac{\delta_{\text{total}}}{L_0} = \frac{0.02}{1} = 0.02 \text{ cm/cm} \quad (1)$$

From the stress-strain diagram, we know that this total strain corresponds to point D and the rod has gone into the **plastic** deformation range. The corresponding stress in the rod upon loading is 480 MPa. Then, according to the definition of stress, we have:

$$P = \sigma_D \cdot A = 480 \times 10^6 \times \frac{\pi}{4} \times 0.02^2 = \boxed{150.80 \text{ kN}}$$

- (b) When the force P is reduced to half of its original value, the stress-strain curve will be along the same slope as defined by Modulus of Elasticity (unloading line: DE) and reach the point S . Since the external force P is reduced to half, the stress is also reduced to half, so point S should be in the middle of DE . Then, the total strain at point S is (taking $\triangle DEF$ for analysis):

$$\epsilon_S = \epsilon_{\text{plastic}} + 1/2 \epsilon_{\text{elastic}} \quad (2)$$

The elastic portion can be determined by using the Hooke's law in $\triangle DEF$:

$$\varepsilon_{\text{elastic}} = \frac{\sigma_D}{E} = \frac{480 \times 10^6}{250 \times 10^9} = 0.00192 \text{ cm/cm} \quad (3a)$$

Taking (3a) into Eq. (1), we get the plastic portion as

$$\varepsilon_{\text{plastic}} = \varepsilon_{\text{total}} - \varepsilon_{\text{elastic}} = 0.02 - 0.00192 = 0.01808 \text{ cm/cm} \quad (3b)$$

Taking (3a) and (3b) into Eq. (2) we get the total strain at point S as:

$$\varepsilon_S = 0.01808 + \frac{1}{2} \times 0.00192 = 0.01904 \text{ cm/cm}$$

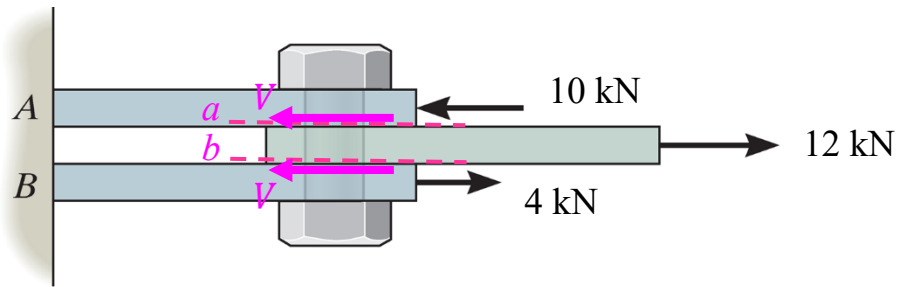
Then, according to the definition of longitudinal strain, the total length of the rod at point S is:

$$L = (1 + \varepsilon_S) \times L_0 = (1 + 0.01904) \times 1 = \boxed{1.01904 \text{ m}}$$

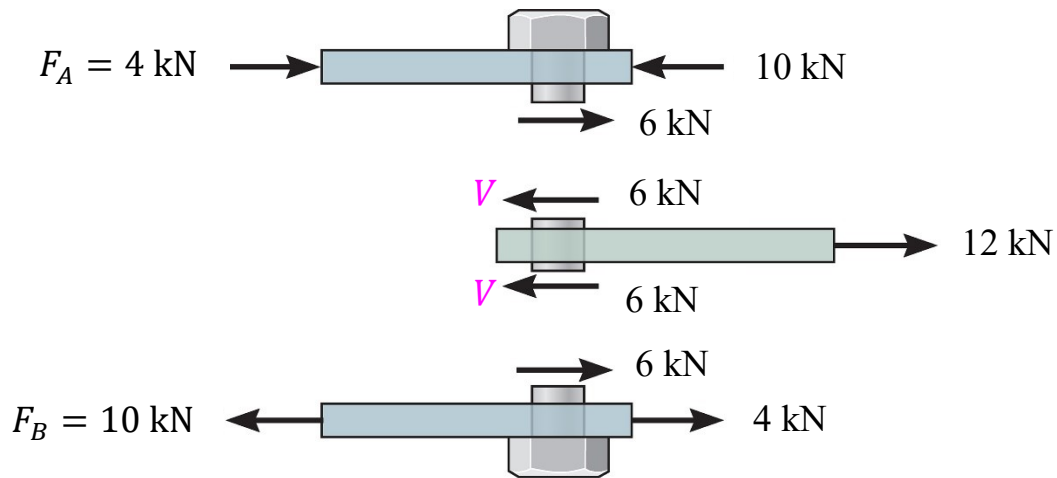
- (c) Continue the process in (b): when the external force P is completely removed, the stress will come back to point E . Now, all elastic strain has been fully recovered (EF part). The total strain remained in the rod is the plastic strain calculated above (OE part), i.e. $\varepsilon_E = \varepsilon_{\text{plastic}} = 0.01808 \text{ cm/cm}$. Thus, the final total length of the rod is:

$$L = (1 + \varepsilon_E) \times L_0 = (1 + 0.01808) \times 1 = \boxed{1.01808 \text{ m}}$$

***[**Bonus Problem**] Determine the internal **shear** forces resisted by the bolt and the resultant force reactions at the point *A* and *B*. (10' in total)



Solutions:



Internal shear force at plane *a* and *b*:

$$V = 12/2 = 6 \text{ kN} \quad (\text{Ans.})$$

Reaction force at point *A*:

$$F_A = 10 - 6 = 4 \text{ kN} \quad (\text{Ans.})$$

Reaction force at point *B*:

$$F_B = 4 + 6 = 10 \text{ kN} \quad (\text{Ans.})$$